

Walking with meaning: cue card

1 Original concept: Tooth brushing

- Bathroom; two minutes, twice a day
- This is all the robot said: instructions
- No room for theory of mind
- Needs a steady grip; if the hands fail, the concept fails

2 Think of your last vacation.

- Ask the room; wait twenty seconds
- Two or three people share
- Then, out loud: "Hands up if you were moving: walking, hiking, swimming, cycling"
- Count the hands

3 Moving helps memory

- Link back to the hands
- Walking AR tour against a seated VR tour of the same museum
- Large effect: $d = 1.31$, 28 adults
- Fresh air and movement also steady emotions

4 More to talk about

- Walks bring residents together
- Side by side, a pause costs less: no face to watch, things to look at
- Less repetition in the final presentation

5 What a walk can serve

- Eight Human Values from our Foundation
- Company: a chat, a mate, less loneliness
- Dignity: a slip is not remarked on
- Truthfulness: no deception; painful news from a trusted person
- Privacy: no secret passed on; nothing leaves the care home
- Identity: the old trade, the old town

6 Walking with meaning

- Walk: fresh air and movement stimulate memory and steady emotions
- Talk: with a fellow resident, or with the guide where agreed
- Ask: consent before joining someone, and before every conversation starter
- Wait: the speaker gets the first chance to fix a slip
- Redirect: gently, to the place, the thread, or a shared topic

7 SA-01 · A walk on your own

- Only residents assessed fit; enclosed garden loop
- Loneliness: some stop going out
- "Is my husband coming?": told bluntly by someone who did not know, or told "He's at work"
- Addressed by: the guide as company or a bridge to a person; painful questions passed to a trusted person

8 SA-02 · A walk with a fellow resident

- Slips: a lost thread, a missing word, a story told twice, a belief from another time
- Both often have dementia: neither notices the other's slips
- Also: two who dislike each other; "I suppose so", going along to please
- Addressed by: asking each alone; a relationship map; the guide keeps the thread

9 Stakeholders

- Seven stakeholders; two walk, three support, two set limits
- The family agrees what may be said, and by whom

10 Problem scenarios

- Four on pair walks, two on walks alone
- In each, something the walk depends on is left to chance: memory, consent, attention, company, trust
- PS-05: a care worker introduces two widows by their grief; it is repeated in front of others

11 Human values

- Social connectedness leads: company is the reason to walk
- Truthfulness: no deception by design; errors corrected; painful truths from a trusted person

12 Value tensions

- Right column: how the design resolves each tension

13 Human factors concepts

- HFC-01 is the vacation question
- HFC-05: theory of mind weakens in dementia; the System keeps a simple model of each walker

14 Measures

- Validated instruments where they exist: OERS, the De Jong Gierveld loneliness scale
- Nothing is recorded: observers code live

15 Evaluation methods

- Detailed in the Evaluation section

16 Technology options

- Six selected, three rejected; the rejections are part of the rationale
- No internet connection: nothing leaves the care home

17 What we keep from morning care

- New activity, same principles
- Graduated prompting became wait, then redirect gently
- The care-worker handover became the trusted person
- Three-item record then, six items now
- Also kept: staged tests, the within-resident comparison, adverse claims

18 Jan

- Our first resident persona
- Postal worker for thirty years, on foot and by bike
- Moderate dementia: today is gone, the round is vivid
- Repeats stories; misses a partner's slip; laughs their own off
- "I'm not a dog to be walked"

19 Elena

- Our second resident persona
- Mild Alzheimer's disease; notices her own slips, and is embarrassed by them
- Widowed last year; lonely since
- Lives two doors from Jan; both worked for the post in the same town
- "Give me a moment first"

20 Thread in the pocket, person in reach

- Pepper at the door: asks each walker alone, introduces, welcomes back
- A pocket guide each; a copy of the System per resident, on a local server
- Beacons at the pond, roses, bench and gate
- Listens on the device; no GPS, no route; nothing leaves the home
- The speaker repairs first, the companion second, the System third

21 What the guide keeps track of

- Theory of mind, kept simple: one model per walker, for one walk
- Common ground: what they share, and what may be mentioned
- The thread, what was said already, a belief from another time, the walker's state
- It decides when to wait, redirect or pass on
- Deleted at the end of the walk

22 How the voice speaks

- Three rules
- Statements, not questions, while walking
- Never "What were you saying?", "You told me", "Do you remember?"

23 Talking on the move

- Talking takes attention from the path; good talk hides tiredness
- Starters only at a stop; no questions while walking
- A second stumble alerts the care worker
- The physiotherapist sets the walk length; the resident can shorten it

24 Is Thomas coming today?

- Elena walks alone; Thomas died last year
- The relationship map lists his death, the trusted person, and the approach agreed with her son
- The guide never lies and never denies: it calls Myra
- Myra comes, sits down, and answers as agreed

25 Settings follow the resident

- The answer to "a dog being walked": the walk serves the resident's own reasons
- Each resident decides how much help, and what may be shared
- Jan: bridge only, no word help. Elena: company on walks alone, word help after a wait

26 What could go wrong

- Help can expose the slip it means to hide
- The guide may cut in before the speaker repairs
- Talking on the move: more stumbles, tiredness missed
- The walk may feel like being walked, for staff's sake
- A listening guide may feel like eavesdropping
- An out-of-date map may let a secret through
- Each one tested; if true, the design changes

27 Design scenarios

- One per walk

28 Personas and robot profiles

- Three people, two devices, one voice
- Settings per resident

29 Objective stories

- Each ties a Function to a value, in a stakeholder's words
- OS-02 is Myra's: "I want the guide to fetch me, not to answer her"

30 Objectives

- Six Musts; talk that recovers comes first

31 Use cases and functions

- Two use cases share six functions
- UC01, the pair walk, is the main use case

32 Claims

- Nine claims, five adverse
- Success criteria in the Evaluation section

33 Design patterns

- One team pattern, three interaction patterns

34 Evaluation

- Verification: do the Functions work as specified (Premises)
- Validation: do they have the intended effect (Claims)
- Four phases, sixteen weeks

35 Evaluation plan

- Phase 1: prototype and staged checks, no residents
- Phase 2: operator-voiced pilot; tune the wait and the wording
- Phase 3: ABAB validation of the Claims, six pairs
- Phase 4: interviews with residents, relatives and staff
- Ethics approval before any resident takes part

36 The prototype: three builds

- Shows every Function end to end
- The screen shows each walker's model live, and why the guide chose each step
- The demo ends by showing what is not kept: no audio file, no traffic outside the home

37 Phase 1: staged verification

- Staged events against a timestamped reference log
- Premises PR1-PR6
- Numbers are proposals; care workers set the final ones

38 Phase 2: operator-voiced pilot

- Formative: before anything is automated
- Each factor varied across short walks
- What residents respond to best becomes the default

39 Phase 3: summative validation

- Single-case ABAB: baseline, intervention, baseline, intervention
- Six pairs of residents, 12 in all; each pair is its own control
- Phase changes on randomly drawn days
- At least 5 observed walks per phase
- Safety alerts stay on in every phase

40 Observed walks: live listening, no recording

- One researcher per walk, about 10 m behind, listening through a headset
- Asked before each walk: may the researcher listen today?
- Emotion rated live (OERS); every slip coded: type, who repaired, how fast, at what cost to face
- Codes only, never words
- Stop at any sign of distress

41 Success criteria

- Each criterion is set before the study starts
- Any disclosure, or an error that causes distress, stops the B phase
- A null result on loneliness is likely; we report it either way

+ Closing, if time: next steps

- Relationship map and consent forms with care staff
- Evidence pass on the references
- Build A of the prototype on the garden loop
- Ethics approval before any resident takes part
- Open for questions